

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	32.7	9.3	2	0.7	0.6	54.7
U3 - Toxicity prior Beta(0.15, 0.85)	37.2	7.4	0.8	0.3	0.2	54.1
U2 - Toxicity prior Beta(0.3, 0.7)	34.6	9.2	2.3	0.7	0.6	52.6
U3 - Toxicity prior Beta(0.3, 0.7)	38.2	8	1	0.2	0.2	52.4
U2 - Toxicity prior Beta(0.5, 0.5)	35.3	9.3	2.3	1.1	0.6	51.4
U3 - Toxicity prior Beta(0.5, 0.5)	38	8.7	1.3	0.4	0.2	51.4
U2 - Toxicity prior Beta(1, 1)	33.8	9.9	3.2	0.9	0.6	51.6
U3 - Toxicity prior Beta(1, 1)	39.2	8.2	1	0.1	0.1	51.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	14.6	4.2	2.5	1.1	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	14.6	4.2	2.6	1.2	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	14.1	4.1	2.9	1.4	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	14.3	4	2.8	1.4	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	14.1	3.9	2.9	1.5	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	14	4	2.9	1.5	0.8	
U2 - Toxicity prior Beta(1, 1)	14.1	3.9	2.9	1.6	0.9	
U3 - Toxicity prior Beta(1, 1)	14	3.9	2.9	1.6	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.7	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	1.9	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	1.9	0.7	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	1.9	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	1.9	0.7	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	1.7	0.6	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	1.7	0.6	0.5	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	34.7	12.7	34.3	11.7	4.7	1.9
U3 - Toxicity prior Beta(0.15, 0.85)	37.3	13.3	32.2	11.9	3.5	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	34.2	11.3	33.9	13.6	5.2	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	36	14	32	11.9	4.3	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	34.3	11.4	33.6	13.5	5.4	1.8
U3 - Toxicity prior Beta(0.5, 0.5)	37.4	11.8	31.4	13.1	4.5	1.8
U2 - Toxicity prior Beta(1, 1)	33.8	11.7	32.6	14.7	5.4	1.8
U3 - Toxicity prior Beta(1, 1)	35.6	13.2	31.6	13.8	4	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.9	4.9	7.2	4.1	2.6	
U3 - Toxicity prior Beta(0.15, 0.85)	11	4.9	7.1	4.3	2.6	
U2 - Toxicity prior Beta(0.3, 0.7)	10.6	4.6	6.9	4.7	2.9	
U3 - Toxicity prior Beta(0.3, 0.7)	10.6	4.6	7	4.7	2.9	
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	4.4	6.8	5	3	
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	4.4	6.8	5	3.1	
U2 - Toxicity prior Beta(1, 1)	10.4	4.4	6.6	5.1	3.3	
U3 - Toxicity prior Beta(1, 1)	10.4	4.4	6.7	5	3.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.3	1.3	0.9	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.3	1.3	1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.2	1.3	1	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.2	1.3	1	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.2	1.3	1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.1	1.3	1	0.4	
U2 - Toxicity prior Beta(1, 1)	0.3	2.1	1.2	1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.3	2.1	1.2	1	0.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.6	34.5	25.7	8.4	3.1	9.7
U3 - Toxicity prior Beta(0.15, 0.85)	20.4	37.7	21.5	8	2.5	9.9
U2 - Toxicity prior Beta(0.3, 0.7)	18.6	34.9	24.5	9.3	3.1	9.6
U3 - Toxicity prior Beta(0.3, 0.7)	20.6	37.3	22.3	7.7	2.5	9.6
U2 - Toxicity prior Beta(0.5, 0.5)	18	33.4	24.7	10.8	3.6	9.5
U3 - Toxicity prior Beta(0.5, 0.5)	21.2	35.9	22.6	8.2	2.6	9.5
U2 - Toxicity prior Beta(1, 1)	18.7	32.1	25	10.2	4.5	9.5
U3 - Toxicity prior Beta(1, 1)	20.2	35.4	23.4	8.8	2.6	9.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	6.9	6.4	3.9	2.5	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	6.9	6.4	4	2.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9	6.4	6.2	4.5	2.8	
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	6.3	6.3	4.5	2.8	
U2 - Toxicity prior Beta(0.5, 0.5)	8.9	6.3	6.1	4.7	3	
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	6.2	6.2	4.7	3	
U2 - Toxicity prior Beta(1, 1)	8.8	6.3	5.9	4.7	3.3	
U3 - Toxicity prior Beta(1, 1)	8.9	6.2	6	4.7	3.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.3	1.1	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.3	1.1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.3	1.2	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.3	1.2	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.3	1.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.2	1.2	0.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.2	1.2	0.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.2	1.2	0.4	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.2	6.3	20.5	39.5	23.6	1.9
U3 - Toxicity prior Beta(0.15, 0.85)	13.3	6.8	18.5	39.1	20.4	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	7.5	5.8	18.7	41.5	24.7	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	12.1	6.4	16.3	42.1	21.2	1.9
U2 - Toxicity prior Beta(0.5, 0.5)	7.5	5.3	17	43.1	25.2	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	11.2	6.5	16.8	41.7	22	1.8
U2 - Toxicity prior Beta(1, 1)	7.1	4.6	17.2	41.6	27.6	1.9
U3 - Toxicity prior Beta(1, 1)	9.7	7.1	15.7	41.4	24.2	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	4	6.2	7.6	7.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4.4	4.1	6.2	7.6	7.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	3.8	6	8	7.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.3	3.7	5.9	8.1	7.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	3.7	5.7	8.1	8	
U3 - Toxicity prior Beta(0.5, 0.5)	4.2	3.7	5.7	8.2	8	
U2 - Toxicity prior Beta(1, 1)	4.2	3.6	5.5	8	8.5	
U3 - Toxicity prior Beta(1, 1)	4.2	3.6	5.4	8	8.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.7	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.7	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	2.2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	2.2	0.8	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.2	29.1	29.9	17.8	13.8	4.2
U3 - Toxicity prior Beta(0.15, 0.85)	9.9	37.4	25.1	15.2	8.2	4.2
U2 - Toxicity prior Beta(0.3, 0.7)	5.2	29.1	28.9	19.7	12.9	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	9.9	34.6	26.4	15.7	9.2	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	4.8	28.9	26.4	21.6	14.1	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	8.7	36.5	24.7	17	8.9	4.2
U2 - Toxicity prior Beta(1, 1)	4.6	30.3	25.4	19.9	15.6	4.2
U3 - Toxicity prior Beta(1, 1)	7.9	36.7	25.5	16.6	9.1	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.8	5.9	6.8	5.6	6.3	
U3 - Toxicity prior Beta(0.15, 0.85)	4.8	5.9	6.7	5.6	6.4	
U2 - Toxicity prior Beta(0.3, 0.7)	4.8	5.7	6.6	5.8	6.6	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	5.6	6.6	5.8	6.6	
U2 - Toxicity prior Beta(0.5, 0.5)	4.8	5.6	6.3	6	6.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.8	5.6	6.3	5.9	6.8	
U2 - Toxicity prior Beta(1, 1)	4.8	5.5	6.3	6	6.9	
U3 - Toxicity prior Beta(1, 1)	4.8	5.5	6.3	5.9	6.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	1.7	2.1	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	1.7	2.1	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	1.8	2.2	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	1.8	2.1	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3.2	1.7	2.2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.2	1.7	2.2	1.2	
U2 - Toxicity prior Beta(1, 1)	0.2	3.1	1.7	2.1	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	3.2	1.8	2.1	1.1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	66.9	12.6	4.8	1.6	0.7	13.4
U3 - Toxicity prior Beta(0.15, 0.85)	69.3	11.8	3.9	1.2	0.4	13.4
U2 - Toxicity prior Beta(0.3, 0.7)	65.3	12.7	6.4	2.1	0.6	12.9
U3 - Toxicity prior Beta(0.3, 0.7)	68	12.6	4.7	1.6	0.2	12.9
U2 - Toxicity prior Beta(0.5, 0.5)	64.9	12.7	6.5	2.3	0.8	12.8
U3 - Toxicity prior Beta(0.5, 0.5)	68.1	12	4.8	2	0.3	12.8
U2 - Toxicity prior Beta(1, 1)	64.7	12.4	6.9	2.2	1	12.8
U3 - Toxicity prior Beta(1, 1)	67.9	12.3	4.9	1.6	0.5	12.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.6	4.1	2.8	1.4	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	18.7	4.1	2.7	1.4	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	18.3	3.8	3.1	1.6	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	18.4	3.8	3.1	1.5	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	18.2	3.9	3	1.6	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	18.2	3.9	3.1	1.6	0.9	
U2 - Toxicity prior Beta(1, 1)	18.1	3.9	3	1.7	0.9	
U3 - Toxicity prior Beta(1, 1)	18.1	3.9	3	1.7	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	1.4	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	1.5	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.4	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.4	0.6	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	1.3	0.5	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	1.3	0.5	0.4	0.1	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.8	31.8	23.4	10.6	5.1	2.3
U3 - Toxicity prior Beta(0.15, 0.85)	28.2	34.8	22.4	8.5	3.8	2.3
U2 - Toxicity prior Beta(0.3, 0.7)	26.4	30.6	22.3	11.8	6.7	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	28.1	34	21.5	9.2	5	2.2
U2 - Toxicity prior Beta(0.5, 0.5)	25.8	30	22.2	12.5	7.3	2.2
U3 - Toxicity prior Beta(0.5, 0.5)	27.1	33.7	21.3	10.2	5.5	2.2
U2 - Toxicity prior Beta(1, 1)	24.9	29.7	22.2	12.9	8.1	2.2
U3 - Toxicity prior Beta(1, 1)	27.3	33.2	21.4	10.2	5.7	2.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.5	6.4	5.9	3.8	3.1	
U3 - Toxicity prior Beta(0.15, 0.85)	10.4	6.5	5.9	3.8	3.1	
U2 - Toxicity prior Beta(0.3, 0.7)	10.1	6.2	5.7	4.2	3.4	
U3 - Toxicity prior Beta(0.3, 0.7)	10.1	6	5.8	4.3	3.5	
U2 - Toxicity prior Beta(0.5, 0.5)	10	6	5.7	4.4	3.6	
U3 - Toxicity prior Beta(0.5, 0.5)	9.9	6	5.7	4.4	3.7	
U2 - Toxicity prior Beta(1, 1)	9.9	6	5.6	4.4	3.8	
U3 - Toxicity prior Beta(1, 1)	9.9	6	5.6	4.4	3.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.5	1.1	1	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.5	1.1	1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.4	1.1	1.1	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.4	1.1	1.1	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.4	1.1	1	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.4	1.1	1.1	0.5	
U2 - Toxicity prior Beta(1, 1)	0.2	2.3	1.1	1	0.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.4	1.1	1	0.4	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	43.5	9.3	2.6	2	0.4	42.2
U3 - Toxicity prior Beta(0.15, 0.85)	45.2	8.7	1.8	1.8	0.3	42.2
U2 - Toxicity prior Beta(0.3, 0.7)	42.4	9.5	3.8	2.5	0.7	41.1
U3 - Toxicity prior Beta(0.3, 0.7)	44.9	8.8	2.6	2.2	0.4	41.1
U2 - Toxicity prior Beta(0.5, 0.5)	41.9	10	4.2	2.4	0.8	40.7
U3 - Toxicity prior Beta(0.5, 0.5)	44.4	9.8	2.4	2	0.7	40.7
U2 - Toxicity prior Beta(1, 1)	42.3	8.8	3.9	2.9	1.3	40.8
U3 - Toxicity prior Beta(1, 1)	43.4	9.4	3.1	2.4	0.9	40.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.5	4	2.3	1	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	16.5	3.9	2.2	1.1	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	16.2	3.7	2.6	1.2	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	16.2	3.7	2.6	1.2	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	16	3.7	2.7	1.4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	16.1	3.6	2.6	1.4	0.7	
U2 - Toxicity prior Beta(1, 1)	16	3.6	2.6	1.4	0.8	
U3 - Toxicity prior Beta(1, 1)	16.1	3.6	2.6	1.4	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.7	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.7	0.5	0.3	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.6	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.6	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	1.5	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	1.5	0.6	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.3	1.4	0.5	0.3	0.1	
U3 - Toxicity prior Beta(1, 1)	0.3	1.4	0.5	0.3	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14	8.4	17	11.4	31	18.2
U3 - Toxicity prior Beta(0.15, 0.85)	19.7	10.9	12.9	11.9	26.4	18.2
U2 - Toxicity prior Beta(0.3, 0.7)	13.1	7.5	16.3	11.6	33.6	17.9
U3 - Toxicity prior Beta(0.3, 0.7)	18.8	9.9	13.2	10.7	29.7	17.7
U2 - Toxicity prior Beta(0.5, 0.5)	12.3	7.5	14.4	12.2	36	17.6
U3 - Toxicity prior Beta(0.5, 0.5)	17.9	8.4	13.6	11.7	30.6	17.8
U2 - Toxicity prior Beta(1, 1)	12.2	6.7	11.9	12.7	38.6	17.9
U3 - Toxicity prior Beta(1, 1)	16.5	8.5	11.8	12.1	33.2	17.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.4	4.2	5.2	4	6.1	
U3 - Toxicity prior Beta(0.15, 0.85)	8.5	4.1	5.2	3.9	6.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.2	3.8	4.9	4.3	6.7	
U3 - Toxicity prior Beta(0.3, 0.7)	8.1	3.8	5	4.3	6.8	
U2 - Toxicity prior Beta(0.5, 0.5)	8	3.5	4.9	4.4	7.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.1	3.6	4.8	4.4	7.2	
U2 - Toxicity prior Beta(1, 1)	7.9	3.6	4.5	4.4	7.6	
U3 - Toxicity prior Beta(1, 1)	7.9	3.6	4.5	4.3	7.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.5	1.3	1.5	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.5	1.3	1.5	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.4	1.3	1.5	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.4	1.3	1.5	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.3	1.3	1.5	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.4	1.3	1.5	0.8	
U2 - Toxicity prior Beta(1, 1)	0.3	2.3	1.2	1.5	0.8	
U3 - Toxicity prior Beta(1, 1)	0.3	2.3	1.2	1.5	0.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.4	28.4	13.8	6.1	3.8	36.5
U3 - Toxicity prior Beta(0.15, 0.85)	12.7	30.4	12.8	4.9	2.9	36.3
U2 - Toxicity prior Beta(0.3, 0.7)	10.7	27.5	15	7	3.9	35.9
U3 - Toxicity prior Beta(0.3, 0.7)	12	29.5	13.9	5.8	2.7	36.1
U2 - Toxicity prior Beta(0.5, 0.5)	10.7	26.5	16.5	6.1	4.4	35.8
U3 - Toxicity prior Beta(0.5, 0.5)	11.4	29.7	14	6	3.4	35.5
U2 - Toxicity prior Beta(1, 1)	9.8	26	16.6	7.2	5.1	35.3
U3 - Toxicity prior Beta(1, 1)	10.6	29.4	14.7	6.4	3.8	35.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11	5.8	4.6	2.5	1.8	
U3 - Toxicity prior Beta(0.15, 0.85)	10.9	5.8	4.5	2.6	1.8	
U2 - Toxicity prior Beta(0.3, 0.7)	10.7	5.4	4.6	2.9	2	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	5.4	4.8	2.9	2	
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	5.2	4.8	3	2.2	
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	5.3	4.7	3	2.2	
U2 - Toxicity prior Beta(1, 1)	10.4	5.2	4.6	3.1	2.3	
U3 - Toxicity prior Beta(1, 1)	10.4	5.3	4.5	3.1	2.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.2	0.9	0.7	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.2	0.9	0.7	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.1	0.9	0.7	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.1	1	0.7	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2	0.9	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2	0.9	0.7	0.2	
U2 - Toxicity prior Beta(1, 1)	0.1	1.9	0.9	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	0.1	1.9	0.9	0.7	0.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	67.6	19.8	4.8	1	0	6.8
U3 - Toxicity prior Beta(0.15, 0.85)	71.9	18.1	2.8	0.6	0	6.6
U2 - Toxicity prior Beta(0.3, 0.7)	68.6	18.7	5.5	0.6	0	6.6
U3 - Toxicity prior Beta(0.3, 0.7)	72.5	17.3	3.1	0.5	0	6.6
U2 - Toxicity prior Beta(0.5, 0.5)	68.1	18.5	6.2	0.5	0	6.7
U3 - Toxicity prior Beta(0.5, 0.5)	71.5	17.2	4	0.7	0	6.6
U2 - Toxicity prior Beta(1, 1)	67.6	18.2	6.8	0.8	0	6.6
U3 - Toxicity prior Beta(1, 1)	72.3	15.5	4.7	0.9	0	6.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.5	5.9	3.8	1.4	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	17.5	5.9	3.8	1.3	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	17.1	5.7	4	1.7	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	17.1	5.7	4	1.8	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	16.8	5.7	4	1.9	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	16.9	5.6	4.1	1.9	0.4	
U2 - Toxicity prior Beta(1, 1)	16.7	5.7	4.1	2	0.5	
U3 - Toxicity prior Beta(1, 1)	16.8	5.7	4.1	2	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.4	0.9	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.4	0.9	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	2.3	0.9	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.3	0.9	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.8	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.9	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	2.1	0.8	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	2.1	0.8	0.5	0.1	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.6	11.4	9.8	32.9	26	3.3
U3 - Toxicity prior Beta(0.15, 0.85)	22.3	11.9	6.4	31.4	24.8	3.2
U2 - Toxicity prior Beta(0.3, 0.7)	15.5	10	9.8	33.6	27.9	3.2
U3 - Toxicity prior Beta(0.3, 0.7)	20.9	10.8	6.3	33.7	25	3.3
U2 - Toxicity prior Beta(0.5, 0.5)	14.2	8.9	9.7	35	29	3.2
U3 - Toxicity prior Beta(0.5, 0.5)	20.4	9.7	6.5	33.4	26.7	3.3
U2 - Toxicity prior Beta(1, 1)	14.5	8.2	8.8	33.8	31.5	3.2
U3 - Toxicity prior Beta(1, 1)	18.7	10.1	6.9	32.5	28.5	3.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.9	4.4	5.1	7	8.2	
U3 - Toxicity prior Beta(0.15, 0.85)	5	4.3	5	7.1	8.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.8	4	4.9	7.1	8.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	4.1	4.9	7.2	8.6	
U2 - Toxicity prior Beta(0.5, 0.5)	4.7	3.9	4.8	7.4	8.8	
U3 - Toxicity prior Beta(0.5, 0.5)	4.7	3.9	4.8	7.3	8.9	
U2 - Toxicity prior Beta(1, 1)	4.7	3.8	4.5	7.2	9.3	
U3 - Toxicity prior Beta(1, 1)	4.6	3.9	4.5	7.3	9.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	1.4	2.2	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	1.3	2.2	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	1.4	2.2	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	1.3	2.2	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.7	1.3	2.2	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.7	1.3	2.2	0.9	
U2 - Toxicity prior Beta(1, 1)	0.3	2.6	1.2	2.1	0.9	
U3 - Toxicity prior Beta(1, 1)	0.3	2.6	1.2	2.1	0.8	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.7	29.2	15.6	6.6	2.6	31.3
U3 - Toxicity prior Beta(0.15, 0.85)	16	31.8	13.1	5.4	2.4	31.3
U2 - Toxicity prior Beta(0.3, 0.7)	14.4	28.6	16.1	7.1	3.1	30.7
U3 - Toxicity prior Beta(0.3, 0.7)	15.5	30.7	14.9	6	2.2	30.7
U2 - Toxicity prior Beta(0.5, 0.5)	13.6	27.6	18.1	6.8	3.6	30.3
U3 - Toxicity prior Beta(0.5, 0.5)	15.9	29.1	15.7	6	3	30.3
U2 - Toxicity prior Beta(1, 1)	13.2	27.8	17.2	7.4	4	30.4
U3 - Toxicity prior Beta(1, 1)	15.2	29.2	15.8	6.1	3.1	30.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.6	5.9	5	2.8	1.8	
U3 - Toxicity prior Beta(0.15, 0.85)	10.6	6	4.9	2.8	1.9	
U2 - Toxicity prior Beta(0.3, 0.7)	10.3	5.6	5	3.3	2.1	
U3 - Toxicity prior Beta(0.3, 0.7)	10.3	5.6	5.1	3.3	2.1	
U2 - Toxicity prior Beta(0.5, 0.5)	10.1	5.4	5	3.5	2.2	
U3 - Toxicity prior Beta(0.5, 0.5)	10.1	5.5	5.1	3.4	2.2	
U2 - Toxicity prior Beta(1, 1)	10	5.4	5	3.5	2.4	
U3 - Toxicity prior Beta(1, 1)	10	5.3	5	3.5	2.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.5	1	0.8	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.5	1	0.8	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.3	1	0.9	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.3	1.1	0.9	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.3	1	0.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.3	1	0.9	0.3	
U2 - Toxicity prior Beta(1, 1)	0.1	2.2	1	0.9	0.3	
U3 - Toxicity prior Beta(1, 1)	0.1	2.2	1	0.9	0.3	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.3	13	12.3	27.3	18.7	7.4
U3 - Toxicity prior Beta(0.15, 0.85)	31.9	12.9	9.7	23.4	14.8	7.3
U2 - Toxicity prior Beta(0.3, 0.7)	21.3	11.3	11.3	28.4	20.6	7.1
U3 - Toxicity prior Beta(0.3, 0.7)	28.7	14.8	9.2	24.4	15.8	7.1
U2 - Toxicity prior Beta(0.5, 0.5)	18.2	12.7	10.2	30.1	21.7	7.1
U3 - Toxicity prior Beta(0.5, 0.5)	27.9	13.7	8.7	27.3	15.3	7.1
U2 - Toxicity prior Beta(1, 1)	18.7	10.4	10.4	30.6	22.8	7.1
U3 - Toxicity prior Beta(1, 1)	27.2	12.8	8.5	26.9	17.6	7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.6	5	5.4	6	6.2	
U3 - Toxicity prior Beta(0.15, 0.85)	6.6	4.9	5.4	6	6.2	
U2 - Toxicity prior Beta(0.3, 0.7)	6.4	4.5	5.2	6.4	6.7	
U3 - Toxicity prior Beta(0.3, 0.7)	6.3	4.5	5.2	6.3	6.8	
U2 - Toxicity prior Beta(0.5, 0.5)	6.2	4.3	5.1	6.6	7	
U3 - Toxicity prior Beta(0.5, 0.5)	6.2	4.3	5	6.6	7	
U2 - Toxicity prior Beta(1, 1)	6.2	4.1	4.9	6.6	7.4	
U3 - Toxicity prior Beta(1, 1)	6.1	4.2	4.9	6.6	7.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3	1.4	2.1	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3	1.4	2.1	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.8	1.4	2.1	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.8	1.5	2.1	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.4	2.1	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.4	2.1	0.9	
U2 - Toxicity prior Beta(1, 1)	0.5	2.7	1.3	2.1	0.9	
U3 - Toxicity prior Beta(1, 1)	0.4	2.7	1.3	2.1	0.9	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	54.1	23.4	10.9	3.6	1	7
U3 - Toxicity prior Beta(0.15, 0.85)	67.1	18.7	5.5	1.4	0.3	7
U2 - Toxicity prior Beta(0.3, 0.7)	55.5	23.5	9	3.8	1.2	7
U3 - Toxicity prior Beta(0.3, 0.7)	65	20.7	5.9	1.2	0.2	7
U2 - Toxicity prior Beta(0.5, 0.5)	54.8	22	11.2	4	1	7
U3 - Toxicity prior Beta(0.5, 0.5)	66.5	19.4	5.3	1.8	0.1	6.9
U2 - Toxicity prior Beta(1, 1)	54.4	22.6	10.1	4.4	1.4	7.1
U3 - Toxicity prior Beta(1, 1)	66	19	5.7	1.9	0.3	7.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.2	6.4	5.8	3.6	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.2	6.5	5.8	3.6	2.2	
U2 - Toxicity prior Beta(0.3, 0.7)	10.9	6.1	5.7	4	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	10.8	6.2	5.7	4.1	2.5	
U2 - Toxicity prior Beta(0.5, 0.5)	10.7	6	5.7	4.4	2.6	
U3 - Toxicity prior Beta(0.5, 0.5)	10.8	6.1	5.6	4.3	2.5	
U2 - Toxicity prior Beta(1, 1)	10.6	5.9	5.5	4.3	2.9	
U3 - Toxicity prior Beta(1, 1)	10.6	5.9	5.5	4.4	2.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.3	1.6	1.5	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.3	1.6	1.4	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.1	1.7	1.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.1	1.6	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.1	1.6	1.6	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	3.1	1.6	1.5	0.5	
U2 - Toxicity prior Beta(1, 1)	1.1	3	1.4	1.5	0.6	
U3 - Toxicity prior Beta(1, 1)	1	3	1.5	1.5	0.5	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	31.6	21.9	17.4	14.9	14.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	39.6	25.8	16.2	11.2	7.1	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	29.8	22.3	18	15.7	14.1	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	39.9	26.1	15.2	11.1	7.6	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	30.4	21.3	17.6	15.6	15	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	39	26.1	14.9	12.2	7.7	0.1
U2 - Toxicity prior Beta(1, 1)	30.4	21.6	17	15.3	15.6	0.1
U3 - Toxicity prior Beta(1, 1)	38.1	26.7	15.5	12.1	7.5	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.2	5.2	5.3	5.6	7.7	
U3 - Toxicity prior Beta(0.15, 0.85)	6.1	5.2	5.4	5.6	7.7	
U2 - Toxicity prior Beta(0.3, 0.7)	6.1	5.2	5.3	5.6	7.8	
U3 - Toxicity prior Beta(0.3, 0.7)	6.2	5.1	5.3	5.7	7.7	
U2 - Toxicity prior Beta(0.5, 0.5)	6.1	5.1	5.3	5.6	7.8	
U3 - Toxicity prior Beta(0.5, 0.5)	6.1	5.2	5.3	5.6	7.8	
U2 - Toxicity prior Beta(1, 1)	6.1	5.1	5.3	5.6	7.9	
U3 - Toxicity prior Beta(1, 1)	6.1	5.1	5.3	5.6	7.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.2	3.1	1.8	2.6	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	3.1	1.8	2.5	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	1.2	3	1.8	2.5	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	3	1.8	2.6	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	3	1.8	2.6	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	3.1	1.8	2.6	1.5	
U2 - Toxicity prior Beta(1, 1)	1.2	3	1.8	2.6	1.5	
U3 - Toxicity prior Beta(1, 1)	1.2	3	1.8	2.5	1.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	34.1	43.6	19.7	2.2	0.1	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	37.3	44.6	16.3	1.4	0.1	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	32.6	45.4	18.8	2.8	0.1	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	35.9	45.3	16.1	2.2	0.2	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	32.6	43.3	20.5	3.3	0	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	34.9	45.5	17.2	2	0.1	0.3
U2 - Toxicity prior Beta(1, 1)	32.2	44.2	18.8	4.5	0	0.3
U3 - Toxicity prior Beta(1, 1)	35.1	45.7	16.1	2.7	0.1	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	9.2	7.7	2.7	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	9.7	9.1	7.8	2.7	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	9.4	8.9	7.7	3.3	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	9.3	9	7.8	3.2	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	9.1	8.9	7.7	3.5	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	9.1	8.9	7.7	3.5	0.7	
U2 - Toxicity prior Beta(1, 1)	9	9	7.6	3.5	0.8	
U3 - Toxicity prior Beta(1, 1)	9	9	7.5	3.5	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	1.2	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.6	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	1.2	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.5	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.5	1.2	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	2.5	1	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	2.5	1	0.6	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.3	7.5	9.3	12.9	59.2	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	16.7	10.9	8.1	11.6	52	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	9.7	8	9	12.9	59.7	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	16.6	10.8	8.2	10.6	53.1	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	9.8	7.1	9.8	12.2	60.4	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	15.9	10.1	8.1	10.6	54.6	0.7
U2 - Toxicity prior Beta(1, 1)	10.4	5.6	9.1	12.5	61.7	0.7
U3 - Toxicity prior Beta(1, 1)	15.1	9.1	8	10.3	56.8	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	3.7	4.5	5.2	12.7	
U3 - Toxicity prior Beta(0.15, 0.85)	3.8	3.8	4.5	5.2	12.7	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	3.7	4.4	5.2	12.9	
U3 - Toxicity prior Beta(0.3, 0.7)	3.8	3.6	4.4	5.1	13	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	3.6	4.4	5.2	12.9	
U3 - Toxicity prior Beta(0.5, 0.5)	3.8	3.6	4.3	5.2	12.9	
U2 - Toxicity prior Beta(1, 1)	3.7	3.6	4.3	5.2	13.1	
U3 - Toxicity prior Beta(1, 1)	3.7	3.6	4.3	5.1	13.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.2	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.2	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.2	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.2	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.2	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.2	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.2	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	2.2	1.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.5	49.2	18	2.2	0	8.1
U3 - Toxicity prior Beta(0.15, 0.85)	25.3	49.9	15	1.8	0	8
U2 - Toxicity prior Beta(0.3, 0.7)	20.6	49.3	18.7	3.3	0	8.1
U3 - Toxicity prior Beta(0.3, 0.7)	25.7	47.9	15.9	2.4	0	8.1
U2 - Toxicity prior Beta(0.5, 0.5)	22	47.7	19.6	2.6	0	8.1
U3 - Toxicity prior Beta(0.5, 0.5)	25.2	48.2	16.5	2	0	8.1
U2 - Toxicity prior Beta(1, 1)	21	47.5	20.3	3.2	0	8
U3 - Toxicity prior Beta(1, 1)	24.4	47.6	17	3	0	8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.6	9.7	7.5	2.7	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	8.5	10	7.4	2.7	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	8.2	9.4	7.7	3.2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	8.2	9.4	7.7	3.2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	8.2	8.9	7.8	3.6	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	8.1	9	7.7	3.6	0.8	
U2 - Toxicity prior Beta(1, 1)	8.1	9.1	7.6	3.6	0.9	
U3 - Toxicity prior Beta(1, 1)	8.1	9	7.5	3.7	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.4	1.3	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.4	1.3	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	1.4	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.3	1.4	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.3	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.3	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	3	1.2	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	3.1	1.2	0.8	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14	41	29.3	9.5	2	4.2
U3 - Toxicity prior Beta(0.15, 0.85)	20	45.1	23.2	6.8	0.7	4.2
U2 - Toxicity prior Beta(0.3, 0.7)	14.3	38.5	29.8	10.3	2.9	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	21.1	42.9	22.6	7.9	1.3	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	14.2	39.8	27.8	11	3	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	20.6	42	22.8	8.8	1.6	4.2
U2 - Toxicity prior Beta(1, 1)	13.8	37.8	28.9	12.2	3.1	4.2
U3 - Toxicity prior Beta(1, 1)	21.5	42.4	22.2	8.1	1.6	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.5	8	8.2	4.9	2.9	
U3 - Toxicity prior Beta(0.15, 0.85)	5.5	8	8.2	4.9	2.8	
U2 - Toxicity prior Beta(0.3, 0.7)	5.5	7.5	8	5.4	3.1	
U3 - Toxicity prior Beta(0.3, 0.7)	5.4	7.6	7.9	5.4	3.2	
U2 - Toxicity prior Beta(0.5, 0.5)	5.4	7.3	7.8	5.7	3.3	
U3 - Toxicity prior Beta(0.5, 0.5)	5.4	7.3	7.8	5.7	3.3	
U2 - Toxicity prior Beta(1, 1)	5.3	7.1	7.7	5.7	3.7	
U3 - Toxicity prior Beta(1, 1)	5.4	7.1	7.6	5.8	3.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.5	1.8	1.7	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.5	1.8	1.7	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.4	1.8	1.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.4	1.8	1.7	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.8	1.7	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.8	1.7	0.6	
U2 - Toxicity prior Beta(1, 1)	0.4	3.3	1.7	1.6	0.6	
U3 - Toxicity prior Beta(1, 1)	0.4	3.3	1.7	1.6	0.6	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	59.1	1.5	0	0	0	39.4
U3 - Toxicity prior Beta(0.15, 0.85)	60.3	1.2	0.1	0	0	38.4
U2 - Toxicity prior Beta(0.3, 0.7)	60.7	1.1	0	0	0	38.2
U3 - Toxicity prior Beta(0.3, 0.7)	60.5	1	0	0	0	38.5
U2 - Toxicity prior Beta(0.5, 0.5)	61	1	0	0	0	38
U3 - Toxicity prior Beta(0.5, 0.5)	61.1	0.9	0	0	0	38
U2 - Toxicity prior Beta(1, 1)	60.2	1.2	0.2	0	0	38.4
U3 - Toxicity prior Beta(1, 1)	61.2	1.1	0.1	0	0	37.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20.4	2.3	0.7	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20.5	2.3	0.7	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	20.3	2.4	0.8	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	20.3	2.4	0.8	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	20.2	2.5	0.9	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	20.2	2.5	0.9	0.2	0	
U2 - Toxicity prior Beta(1, 1)	20.1	2.6	0.9	0.2	0	
U3 - Toxicity prior Beta(1, 1)	20.1	2.6	0.9	0.2	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	0.6	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.1	0.6	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	28.1	51	13.8	2.7	0.1	4.3
U3 - Toxicity prior Beta(0.15, 0.85)	32.6	49.9	11.1	2	0	4.4
U2 - Toxicity prior Beta(0.3, 0.7)	28	49.2	15.2	3	0.1	4.5
U3 - Toxicity prior Beta(0.3, 0.7)	32.3	48.8	11.4	3	0	4.5
U2 - Toxicity prior Beta(0.5, 0.5)	29.2	46.6	17	2.9	0.1	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	35.1	46.1	12.1	2.5	0	4.2
U2 - Toxicity prior Beta(1, 1)	28.6	48.3	15.3	3.6	0	4.2
U3 - Toxicity prior Beta(1, 1)	34.5	46.6	11.5	3.2	0	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.5	11.5	6.8	2.2	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	8.4	11.5	6.8	2.2	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	8.2	11	7	2.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	8.1	11.2	7	2.7	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	8.1	10.6	7	3.2	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	8	10.5	7.1	3.2	0.7	
U2 - Toxicity prior Beta(1, 1)	7.9	10.5	7	3.3	0.8	
U3 - Toxicity prior Beta(1, 1)	7.9	10.5	7	3.3	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.3	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.3	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.6	1.3	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.7	1.3	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	3.5	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	1.2	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.4	1.1	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.4	1.1	0.6	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.5	4.1	0.3	0	0	73.1
U3 - Toxicity prior Beta(0.15, 0.85)	22	3.8	0.2	0	0	74
U2 - Toxicity prior Beta(0.3, 0.7)	23.2	3.3	0.4	0	0	73.1
U3 - Toxicity prior Beta(0.3, 0.7)	23	3.5	0.1	0	0	73.4
U2 - Toxicity prior Beta(0.5, 0.5)	23.3	3.9	0.4	0	0	72.4
U3 - Toxicity prior Beta(0.5, 0.5)	23.5	3.9	0	0	0	72.6
U2 - Toxicity prior Beta(1, 1)	22.8	3.7	0.4	0	0	73.1
U3 - Toxicity prior Beta(1, 1)	23.4	3.4	0	0	0	73.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.5	3.4	1.1	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	16.6	3.3	1.1	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	16.6	3.3	1.2	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	16.7	3.3	1.2	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	16.6	3.3	1.3	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	16.7	3.2	1.3	0.1	0	
U2 - Toxicity prior Beta(1, 1)	16.5	3.4	1.3	0.2	0	
U3 - Toxicity prior Beta(1, 1)	16.5	3.4	1.3	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	1.6	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	1.5	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.5	0.2	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.4	0.3	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	1.4	0.2	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	1.4	0.2	0	0	
U2 - Toxicity prior Beta(1, 1)	0.2	1.2	0.2	0	0	
U3 - Toxicity prior Beta(1, 1)	0.2	1.2	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	1.4	4.9	41.2	51.4	0
U3 - Toxicity prior Beta(0.15, 0.85)	3.2	2.7	5.6	42.3	46.2	0
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	1.3	4.8	41.3	51.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	2.9	2.5	6	42	46.6	0
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	1.3	4.5	41.3	51.7	0
U3 - Toxicity prior Beta(0.5, 0.5)	3.1	2.3	6.1	41.5	47	0
U2 - Toxicity prior Beta(1, 1)	1.4	1.4	4.6	39.7	52.9	0
U3 - Toxicity prior Beta(1, 1)	2.6	2.2	6.4	42.4	46.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.2	3.7	7.8	12	
U3 - Toxicity prior Beta(0.15, 0.85)	3.3	3.2	3.7	7.8	12	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.1	3.7	7.7	12.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.3	3.2	3.7	7.8	12.1	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.2	3.7	7.8	12.1	
U3 - Toxicity prior Beta(0.5, 0.5)	3.3	3.2	3.7	7.7	12.2	
U2 - Toxicity prior Beta(1, 1)	3.3	3.2	3.7	7.7	12.2	
U3 - Toxicity prior Beta(1, 1)	3.3	3.2	3.7	7.7	12.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3	1.2	
U2 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3	1.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3	1.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	36.6	26.2	32.6	3.7	0.2	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	41.4	26.4	28	3.5	0	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	35.7	25.1	33	5.3	0.2	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	41.4	26.5	27	4.3	0.1	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	35.5	25.8	32.2	5.7	0.1	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	41	27	26.5	4.6	0.2	0.7
U2 - Toxicity prior Beta(1, 1)	37.2	23.9	32.1	6.1	0	0.7
U3 - Toxicity prior Beta(1, 1)	41.9	26	26.7	4.6	0.1	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9	7.5	9.4	3.2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	9	7.5	9.5	3.3	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	7.1	9.2	4	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	8.8	7.2	9.2	4	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	7	9.1	4.1	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	8.7	7.1	9.1	4.1	0.9	
U2 - Toxicity prior Beta(1, 1)	8.6	6.9	9.2	4.2	1	
U3 - Toxicity prior Beta(1, 1)	8.6	7	9.2	4.2	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3	1.6	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3	1.6	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.9	1.6	1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2.9	1.6	1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.8	1.5	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.8	1.5	1	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	2.7	1.4	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	2.7	1.4	0.9	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	30.8	24.7	28.9	5.7	0	9.9
U3 - Toxicity prior Beta(0.15, 0.85)	35.2	24.6	25.7	4.5	0.1	9.9
U2 - Toxicity prior Beta(0.3, 0.7)	31	23.6	29.2	6.4	0.1	9.7
U3 - Toxicity prior Beta(0.3, 0.7)	35.9	23.2	24.4	6.8	0	9.7
U2 - Toxicity prior Beta(0.5, 0.5)	30.8	23.5	28.7	7.4	0	9.6
U3 - Toxicity prior Beta(0.5, 0.5)	34.4	23.5	25.2	7.2	0	9.7
U2 - Toxicity prior Beta(1, 1)	32.4	21.6	27.6	8.4	0.2	9.8
U3 - Toxicity prior Beta(1, 1)	35.9	22.6	24.5	7.2	0.1	9.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.5	6.3	8.2	3.2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	10.4	6.3	8.3	3.3	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	10.1	6.1	8.2	3.9	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	10.2	6	8.2	3.9	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	10	5.9	8.1	4.2	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	10.1	5.8	8.1	4.2	0.9	
U2 - Toxicity prior Beta(1, 1)	10	5.8	7.9	4.3	1	
U3 - Toxicity prior Beta(1, 1)	10	5.8	8	4.2	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.8	1.3	1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.8	1.3	1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.7	1.3	1.1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.7	1.3	1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.6	1.3	1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.7	1.3	1	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.6	1.2	1	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	2.5	1.2	1	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.4	22.1	41.2	8.9	0.9	5.5
U3 - Toxicity prior Beta(0.15, 0.85)	25.1	22.6	37.7	8.6	0.6	5.4
U2 - Toxicity prior Beta(0.3, 0.7)	21.8	17.7	44	10.2	0.9	5.4
U3 - Toxicity prior Beta(0.3, 0.7)	26.5	19.9	39.1	8.4	0.7	5.4
U2 - Toxicity prior Beta(0.5, 0.5)	22	17.3	43.9	10.5	1	5.3
U3 - Toxicity prior Beta(0.5, 0.5)	25.7	19	40.2	8.7	1.1	5.3
U2 - Toxicity prior Beta(1, 1)	23	17.2	41.6	11.4	1.4	5.4
U3 - Toxicity prior Beta(1, 1)	26.6	18.2	39.5	9.5	1	5.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8	6.1	9.2	4.5	1.7	
U3 - Toxicity prior Beta(0.15, 0.85)	7.9	5.9	9.4	4.5	1.8	
U2 - Toxicity prior Beta(0.3, 0.7)	7.8	5.6	9.1	5.1	1.9	
U3 - Toxicity prior Beta(0.3, 0.7)	7.8	5.6	9.1	5.1	1.9	
U2 - Toxicity prior Beta(0.5, 0.5)	7.7	5.3	9	5.4	2.1	
U3 - Toxicity prior Beta(0.5, 0.5)	7.6	5.3	9	5.4	2	
U2 - Toxicity prior Beta(1, 1)	7.6	5.3	8.7	5.6	2.2	
U3 - Toxicity prior Beta(1, 1)	7.6	5.3	8.7	5.5	2.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.8	1.6	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	1.6	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	1.7	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	1.7	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.6	1.6	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.6	1.2	0.3	
U2 - Toxicity prior Beta(1, 1)	0.4	2.6	1.5	1.2	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	2.6	1.5	1.2	0.3	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	23.6	36.2	7.5	0.4	0	32.3
U3 - Toxicity prior Beta(0.15, 0.85)	24.8	36.1	6.4	0.4	0	32.3
U2 - Toxicity prior Beta(0.3, 0.7)	24.2	35	7.8	0.6	0	32.4
U3 - Toxicity prior Beta(0.3, 0.7)	25.2	36.3	5.7	0.5	0	32.3
U2 - Toxicity prior Beta(0.5, 0.5)	25	34.5	7.5	0.8	0	32.2
U3 - Toxicity prior Beta(0.5, 0.5)	25.8	35	6.4	0.6	0	32.2
U2 - Toxicity prior Beta(1, 1)	25.4	34.8	7.9	0.8	0	31.1
U3 - Toxicity prior Beta(1, 1)	26.8	34.7	6.4	0.9	0	31.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.6	8.6	4.7	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	11.5	8.6	4.6	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	11.3	8.3	4.9	1.6	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	11.3	8.4	4.9	1.6	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	11.1	7.9	5.3	1.8	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	11.1	8	5.3	1.8	0.3	
U2 - Toxicity prior Beta(1, 1)	10.9	8	5.2	1.9	0.4	
U3 - Toxicity prior Beta(1, 1)	10.9	8.1	5.1	1.9	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	0.9	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	0.9	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.6	0.9	0.4	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.6	0.9	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.5	0.9	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.5	0.9	0.4	0	
U2 - Toxicity prior Beta(1, 1)	0.3	2.3	0.8	0.4	0	
U3 - Toxicity prior Beta(1, 1)	0.3	2.3	0.8	0.4	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	70.8	22	4.4	0.2	0	2.6
U3 - Toxicity prior Beta(0.15, 0.85)	72.6	20.8	3.6	0.3	0	2.7
U2 - Toxicity prior Beta(0.3, 0.7)	70.6	21.3	5.1	0.4	0	2.6
U3 - Toxicity prior Beta(0.3, 0.7)	72.6	20.5	4	0.3	0	2.6
U2 - Toxicity prior Beta(0.5, 0.5)	70.7	21.8	4.8	0.4	0	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	73.2	20.4	3.6	0.4	0	2.4
U2 - Toxicity prior Beta(1, 1)	69.9	22.4	5.1	0.3	0	2.3
U3 - Toxicity prior Beta(1, 1)	72.6	21	3.8	0.3	0	2.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.7	7.2	3.8	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	17.7	7.2	3.8	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	17.3	7	4.1	1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	17.3	7.1	4.1	1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	17	7.3	4.1	1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	17	7.3	4.1	1.1	0.1	
U2 - Toxicity prior Beta(1, 1)	16.9	7.4	4.1	1.1	0.1	
U3 - Toxicity prior Beta(1, 1)	16.9	7.3	4.1	1.1	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2	0.8	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2	0.8	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	1.9	0.8	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	1.9	0.8	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	1.9	0.8	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	1.9	0.7	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.6	1.8	0.7	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.6	1.8	0.7	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.7	12.9	15.4	35.2	13	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	29.4	15.7	12.1	32.1	9.9	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	21.4	11.6	14	37.4	14.8	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	27.9	14.5	11.8	34.2	10.8	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	21.9	11.6	15.1	35	15.6	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	27.6	14.2	10.9	33.6	12.9	0.8
U2 - Toxicity prior Beta(1, 1)	21.5	10.7	13.8	36.1	17.1	0.8
U3 - Toxicity prior Beta(1, 1)	28	13.7	10.6	32.3	14.6	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5	4.2	5.5	8	7.2	
U3 - Toxicity prior Beta(0.15, 0.85)	5.1	4.2	5.5	8	7.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	4.1	5.4	8.2	7.4	
U3 - Toxicity prior Beta(0.3, 0.7)	4.9	4.1	5.4	8.2	7.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	4	5.3	8.2	7.5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	4	5.3	8.2	7.5	
U2 - Toxicity prior Beta(1, 1)	4.9	4	5.1	8.2	7.8	
U3 - Toxicity prior Beta(1, 1)	4.9	4	5.1	8.2	7.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	1.5	2.3	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	1.5	2.3	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.6	1.5	2.3	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.6	1.5	2.3	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.6	1.5	2.2	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.6	1.5	2.3	1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.6	1.4	2.2	0.9	
U3 - Toxicity prior Beta(1, 1)	0.3	2.6	1.4	2.2	0.9	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.6	30.5	27.3	25.9	7.6	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	10.5	38.1	26.7	20	4.6	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	8.2	31.6	28.7	23.7	7.7	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	11	38.6	24.6	20.9	4.8	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	8	30.3	28.4	25.2	8	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	10.4	36.8	27.1	21	4.6	0.1
U2 - Toxicity prior Beta(1, 1)	8.7	30.8	27	24.8	8.6	0.1
U3 - Toxicity prior Beta(1, 1)	10.7	37.9	25.9	20.1	5.3	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	5.4	6.5	7.5	6.4	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	5.4	6.5	7.5	6.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	5.2	6.4	7.6	6.6	
U3 - Toxicity prior Beta(0.3, 0.7)	4.1	5.3	6.3	7.6	6.6	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	5.3	6.5	7.5	6.6	
U3 - Toxicity prior Beta(0.5, 0.5)	4.1	5.3	6.4	7.5	6.7	
U2 - Toxicity prior Beta(1, 1)	4.1	5.2	6.3	7.5	6.8	
U3 - Toxicity prior Beta(1, 1)	4.1	5.3	6.3	7.5	6.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	1.8	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	1.8	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	1.8	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	1.8	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.8	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	1.8	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.4	1.7	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.4	1.7	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.3	12.9	34.6	27.1	14.5	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	16.7	17.1	35.1	22.2	8.3	0.6
U2 - Toxicity prior Beta(0.3, 0.7)	10	13.6	34.8	26.8	14.2	0.6
U3 - Toxicity prior Beta(0.3, 0.7)	17.3	16.8	32.9	22.6	9.8	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	9.9	11.6	36.5	26.9	14.5	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	15.6	16.6	34.2	22.8	10.2	0.6
U2 - Toxicity prior Beta(1, 1)	9.9	11.8	35.2	25.7	16.8	0.6
U3 - Toxicity prior Beta(1, 1)	14.8	17.1	33.8	22.1	11.6	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	4.1	7.4	6.8	7.5	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	4.1	7.4	6.8	7.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	4	7.4	6.8	7.6	
U3 - Toxicity prior Beta(0.3, 0.7)	4.1	4	7.3	6.9	7.6	
U2 - Toxicity prior Beta(0.5, 0.5)	4	4	7.2	7	7.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4	4	7.3	6.9	7.7	
U2 - Toxicity prior Beta(1, 1)	4	3.9	7.2	7	7.8	
U3 - Toxicity prior Beta(1, 1)	4	4	7.2	6.9	7.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	2	2.5	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	2	2.5	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	2	2.5	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.8	2	2.5	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	2	2.5	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	2	2.5	1.2	
U2 - Toxicity prior Beta(1, 1)	0.4	2.8	2	2.5	1.2	
U3 - Toxicity prior Beta(1, 1)	0.4	2.8	2	2.5	1.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	4.2	12.6	38.1	35.8	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	14.8	6.6	12.1	35.3	31.2	0
U2 - Toxicity prior Beta(0.3, 0.7)	8.9	3.8	13.1	36.3	37.9	0
U3 - Toxicity prior Beta(0.3, 0.7)	13.4	6.1	12.2	36.5	31.8	0
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	3.3	12.6	36.9	38.9	0
U3 - Toxicity prior Beta(0.5, 0.5)	13	5.2	11.9	36.5	33.4	0
U2 - Toxicity prior Beta(1, 1)	8.6	3.6	11.7	35.1	41	0
U3 - Toxicity prior Beta(1, 1)	12.8	5.5	11.1	36.1	34.5	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	3.5	4.9	7.8	10.3	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.5	5	7.7	10.3	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	3.4	4.8	7.7	10.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.4	4.9	7.8	10.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.5	3.4	4.7	7.7	10.7	
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	3.4	4.7	7.7	10.7	
U2 - Toxicity prior Beta(1, 1)	3.5	3.4	4.7	7.6	10.9	
U3 - Toxicity prior Beta(1, 1)	3.5	3.4	4.7	7.6	10.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.7	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.7	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.7	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.7	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.5	1.4	2.7	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.5	1.4	2.7	1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.4	2.6	1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	2.6	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.8	12.2	46.9	19	6.1	2
U3 - Toxicity prior Beta(0.15, 0.85)	21.4	14.7	41.8	16.2	4	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	14.9	11.4	43.8	20.2	7.8	1.9
U3 - Toxicity prior Beta(0.3, 0.7)	21.6	14.8	40.2	16.8	4.8	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	15.1	11	43.2	19.8	9	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	22.8	13.7	39.3	16.7	5.7	1.8
U2 - Toxicity prior Beta(1, 1)	14.3	12	41.7	20.3	9.9	1.8
U3 - Toxicity prior Beta(1, 1)	21.6	13.5	38.2	18.1	6.7	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.2	5.4	9.3	5.6	4.3	
U3 - Toxicity prior Beta(0.15, 0.85)	5.2	5.3	9.4	5.6	4.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	5.1	9	6	4.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4.9	5.1	9.1	6.1	4.7	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	4.9	8.8	6.3	4.9	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	5	8.7	6.3	4.9	
U2 - Toxicity prior Beta(1, 1)	4.8	4.7	8.6	6.4	5.2	
U3 - Toxicity prior Beta(1, 1)	4.9	4.7	8.6	6.3	5.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.1	1.9	1.7	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.9	1.7	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.9	1.8	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3	1.9	1.8	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.9	1.7	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.9	1.9	1.8	0.7	
U2 - Toxicity prior Beta(1, 1)	0.4	2.9	1.8	1.7	0.7	
U3 - Toxicity prior Beta(1, 1)	0.4	2.8	1.8	1.7	0.7	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.5	36.1	35.6	10.6	5.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	17.3	41.1	29.1	9.4	3	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	12	36.9	33.2	12.5	5.3	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	17.5	41.7	27.5	10.5	2.7	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	12	36.1	32.4	13.5	5.9	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	17.2	40.7	27.3	11.1	3.6	0.1
U2 - Toxicity prior Beta(1, 1)	12.5	35.3	32	13.7	6.4	0.1
U3 - Toxicity prior Beta(1, 1)	15.9	40.5	28.2	11.5	3.8	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	7.7	8.8	5.3	3.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	7.7	8.8	5.3	3.6	
U2 - Toxicity prior Beta(0.3, 0.7)	4.6	7.3	8.5	5.7	3.9	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	7.4	8.5	5.6	3.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.5	7.2	8.3	6	4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.6	7.2	8.4	5.9	3.9	
U2 - Toxicity prior Beta(1, 1)	4.5	7.1	8.1	6.1	4.2	
U3 - Toxicity prior Beta(1, 1)	4.5	7.1	8.2	6.1	4.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3	1.6	1.4	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3	1.6	1.4	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.6	1.4	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.6	1.4	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.6	1.4	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.6	1.5	0.6	
U2 - Toxicity prior Beta(1, 1)	0.3	2.9	1.5	1.4	0.6	
U3 - Toxicity prior Beta(1, 1)	0.4	2.9	1.5	1.4	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.8	35.2	16.8	7.8	3.1	14.3
U3 - Toxicity prior Beta(0.15, 0.85)	28	36.2	13.3	6.2	1.9	14.4
U2 - Toxicity prior Beta(0.3, 0.7)	22.3	33.3	16.5	9.9	4.1	13.9
U3 - Toxicity prior Beta(0.3, 0.7)	27.6	35.3	12.9	7.7	2.5	14
U2 - Toxicity prior Beta(0.5, 0.5)	20.8	32.4	18.5	9.4	5.1	13.8
U3 - Toxicity prior Beta(0.5, 0.5)	26.6	34.5	14.3	7.3	3.6	13.7
U2 - Toxicity prior Beta(1, 1)	20.5	31.8	16.7	11.4	5.6	14
U3 - Toxicity prior Beta(1, 1)	25.9	33	14.5	9.2	3.7	13.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	8	6	3.3	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9	7.8	6	3.2	2.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.3	7.5	6.1	3.9	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.4	7.5	6.2	3.8	2.5	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	7	6.2	4.3	2.6	
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	7	6.3	4.1	2.7	
U2 - Toxicity prior Beta(1, 1)	8.2	7	6	4.2	3	
U3 - Toxicity prior Beta(1, 1)	8.1	7	6.1	4.2	3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.4	1.3	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.4	1.2	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.3	1.2	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.3	1.2	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.1	1.3	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.1	1.3	1.2	0.4	
U2 - Toxicity prior Beta(1, 1)	0.4	2.9	1.2	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	3	1.2	1.1	0.3	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	30.9	42.9	14.6	3.8	1.2	6.6
U3 - Toxicity prior Beta(0.15, 0.85)	36.2	44.2	9.6	2.9	0.6	6.5
U2 - Toxicity prior Beta(0.3, 0.7)	26.8	43.7	16.1	5.2	1.5	6.7
U3 - Toxicity prior Beta(0.3, 0.7)	35.7	43.4	10.6	2.9	0.8	6.6
U2 - Toxicity prior Beta(0.5, 0.5)	27.5	44.2	15.7	4.4	2	6.2
U3 - Toxicity prior Beta(0.5, 0.5)	35.2	41.8	12.1	3.7	0.8	6.4
U2 - Toxicity prior Beta(1, 1)	28.2	41	17.6	5.3	1.8	6.1
U3 - Toxicity prior Beta(1, 1)	35.9	41	12.4	3.4	1	6.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8	10	6.7	2.8	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	8	10.1	6.5	2.8	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	7.8	9.4	6.8	3.4	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	7.7	9.5	6.8	3.4	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	7.6	9.2	6.8	3.7	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	7.6	9.1	6.8	3.6	1.9	
U2 - Toxicity prior Beta(1, 1)	7.4	9.2	6.6	3.8	2.1	
U3 - Toxicity prior Beta(1, 1)	7.5	9.2	6.6	3.8	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.4	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.4	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.8	1.4	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.8	1.4	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.6	1.4	1.1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.6	1.3	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.6	3.4	1.2	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.6	3.5	1.2	1.1	0.3	